Machine Learning in Epidemiology with R – Part 2

From Trees to Neural Networks



Federica Gazzelloni

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Two-hour online, hands-on R workshop

01 Learning objectives and session route

Understand the data and define the task

Fit a tree and examine pruning

Explore a network through saved results

Evaluate predictions and thresholds

Discuss limitations and next steps

By the end: explain the fit, read the errors and justify an evaluation choice.

02 West Nile surveillance and heart disease

Part 1	Part 2
West Nile surveillance	Clinical measurements
Define a prediction unit and outcome	One patient and existing disease
Preparation and evaluation	The same workflow, new learners

The setting changes. The need to define the task and protect evaluation remains.

03

Meet the heart-disease data

270
patients

13
original predictors

2
outcome classes

heart_disease: 150 absent, 120 present

Examples: age, chest pain, cholesterol, exercise measurements

Checkpoint: What does one row represent? Which column is the outcome?

04 The classification task

Clinical measurements are the predictors.

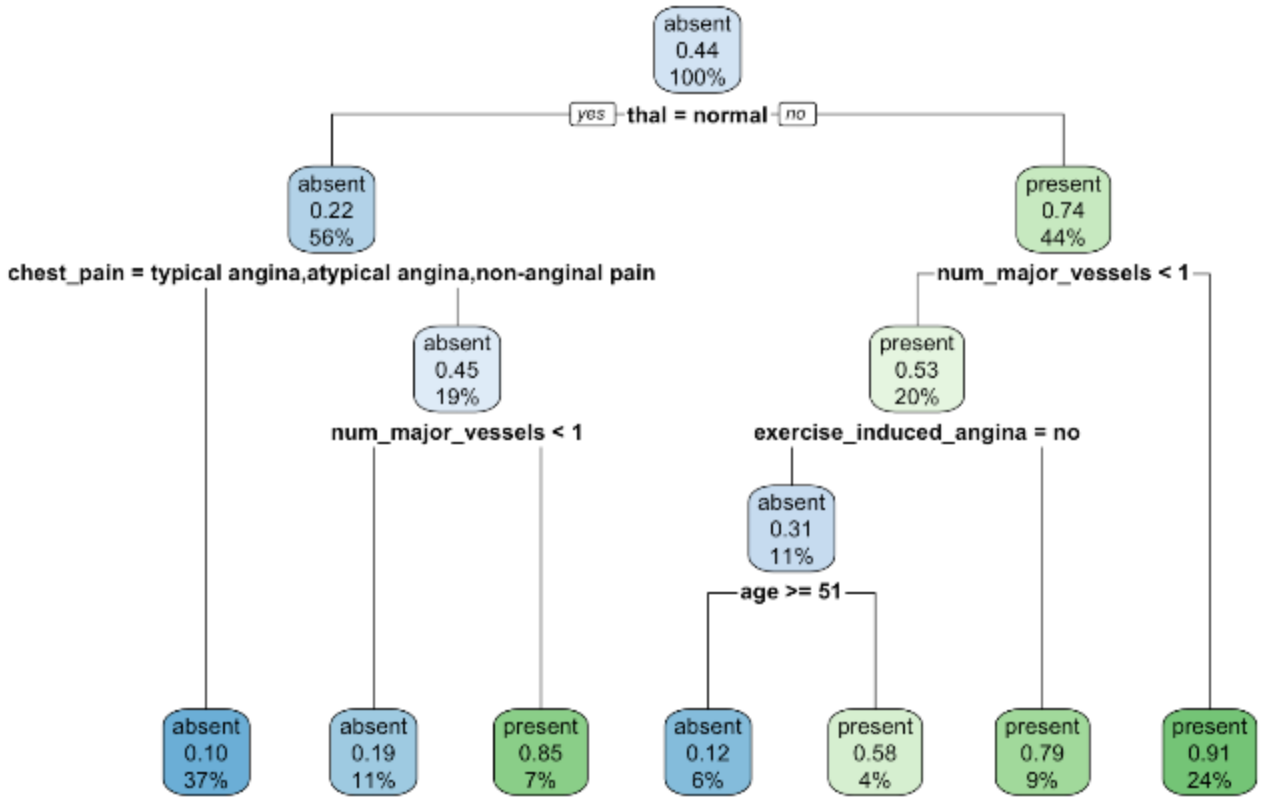
heart_disease is the outcome.

“present” is the positive class.

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target = "heart_disease", positive = "present"
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We classify existing disease. This example does not predict future cardiovascular risk.

How a classification tree splits patients



A split asks a question about one predictor.

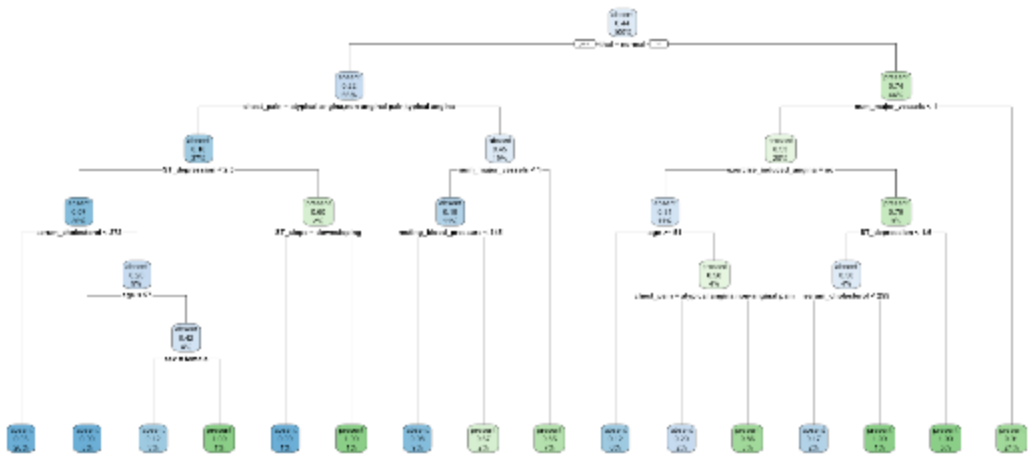
Follow a branch until you reach a leaf.

A leaf gives a class and fitted proportions.

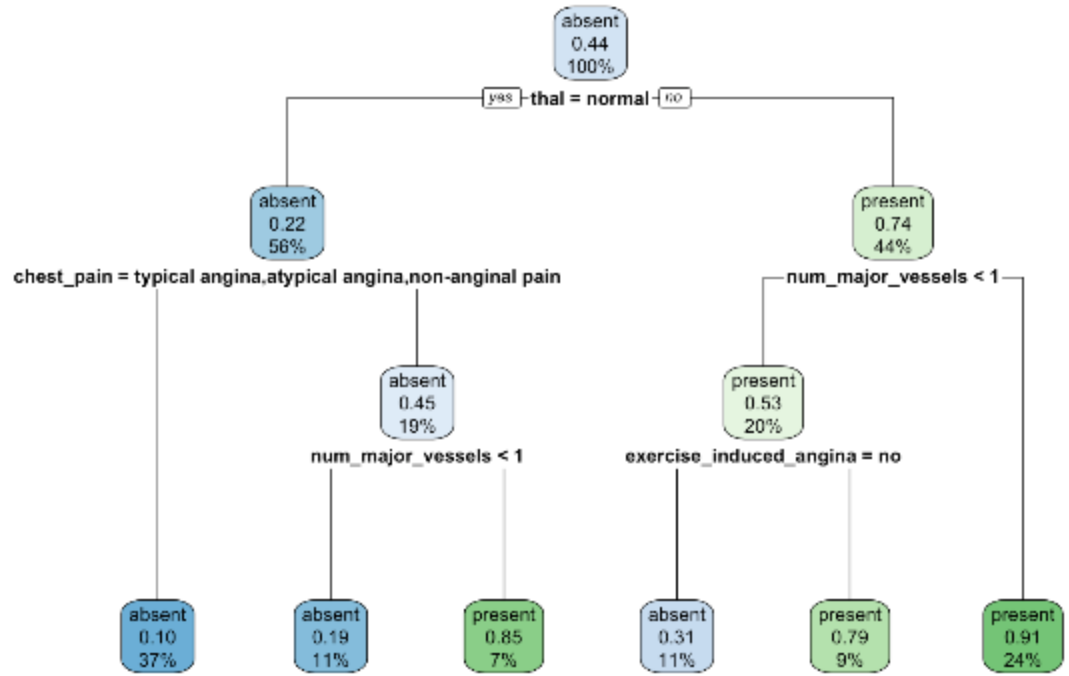
Default tree: fitted on all 270 patients to illustrate learning.

06 Deep and pruned trees

Deep tree: minsplit = 2



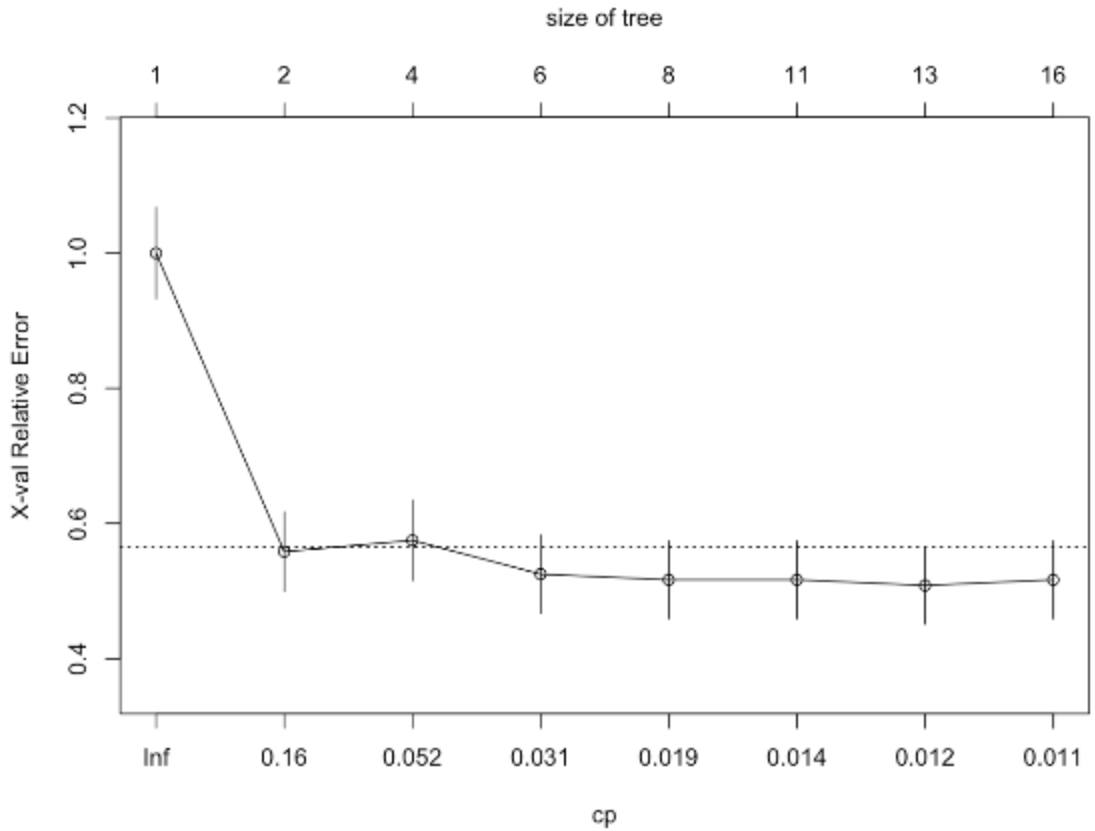
Pruned tree: cp = 0.031



Compare both with the default tree. More branches can fit details of this sample.

07

Overfitting and pruning



Overfitting learns sample details that may not repeat.

Pruning removes branches to limit complexity.

cp = 0.031 retains the article's replication setting.

Checkpoint: Why might pruning help predictions on new patients?

A neural network learns weighted combinations

Inputs	Hidden layers	Output
Encoded measurements	Weighted combinations	Probability of present
Numbers the network receives	ReLU transforms learned signals	Sigmoid maps to 0–1

Training adjusts weights to reduce a loss: a measure of prediction error.

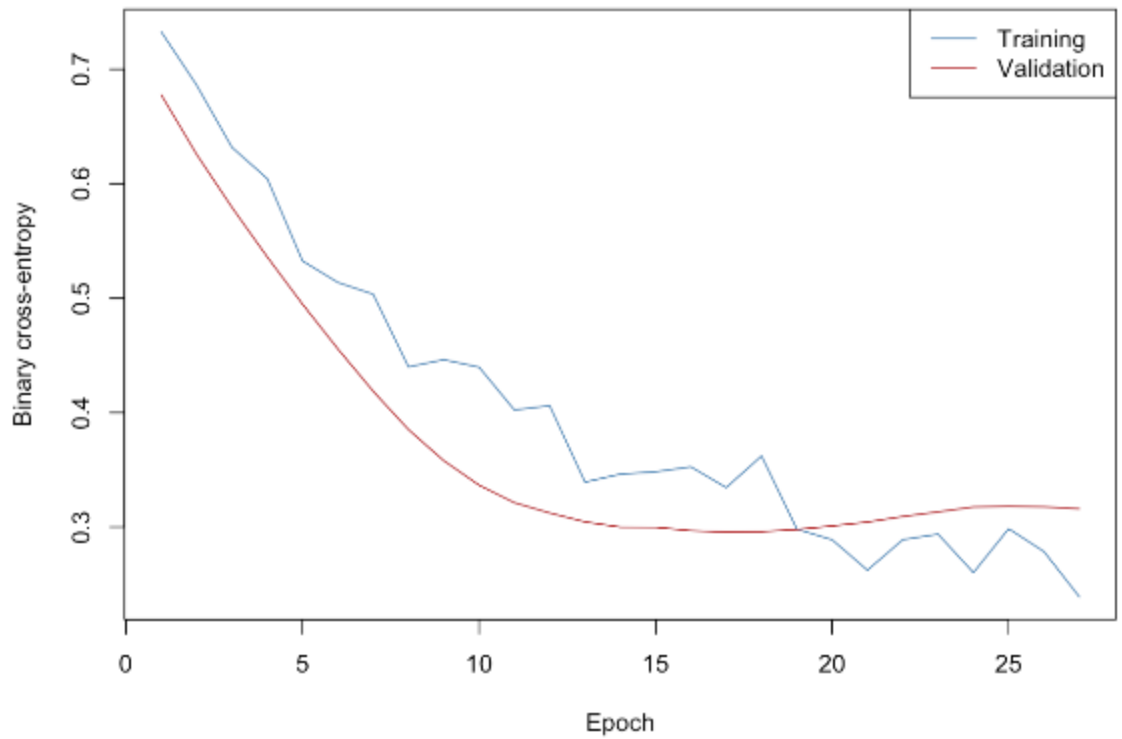
13 predictors become 22 encoded inputs

Input	Hidden layer 1	Hidden layer 2	Output
22 columns	64 units, ReLU	32 units, ReLU	1 unit, sigmoid
6 numeric + 3 binary + 13 indicators	Dropout 0.4	Dropout 0.4	P(disease present)

Dropout temporarily omits 40% of units during training.

Encoding is an explicit local reconstruction. The article does not specify its implementation.

Training and validation loss: local run



Training loss measures fit on rows used for learning.

Validation loss monitors rows kept out of updates.

Early stopping restores the best validation weights.

Checkpoint: What does falling training loss with rising validation loss suggest?

Random-forest holdout evaluation

270 patients

180 training observations	90 test observations
Fit the random forest	Predict held-out patients
Learner default tree count	Read probabilities and errors

Test observations stay outside model fitting and model selection.

The confusion matrix records consequences

Observed outcome	Predicted present	Predicted absent
Present	30 true positives	7 false negatives
Absent	8 false positives	45 true negatives

False positive: predicts disease for a patient whose outcome is absent.

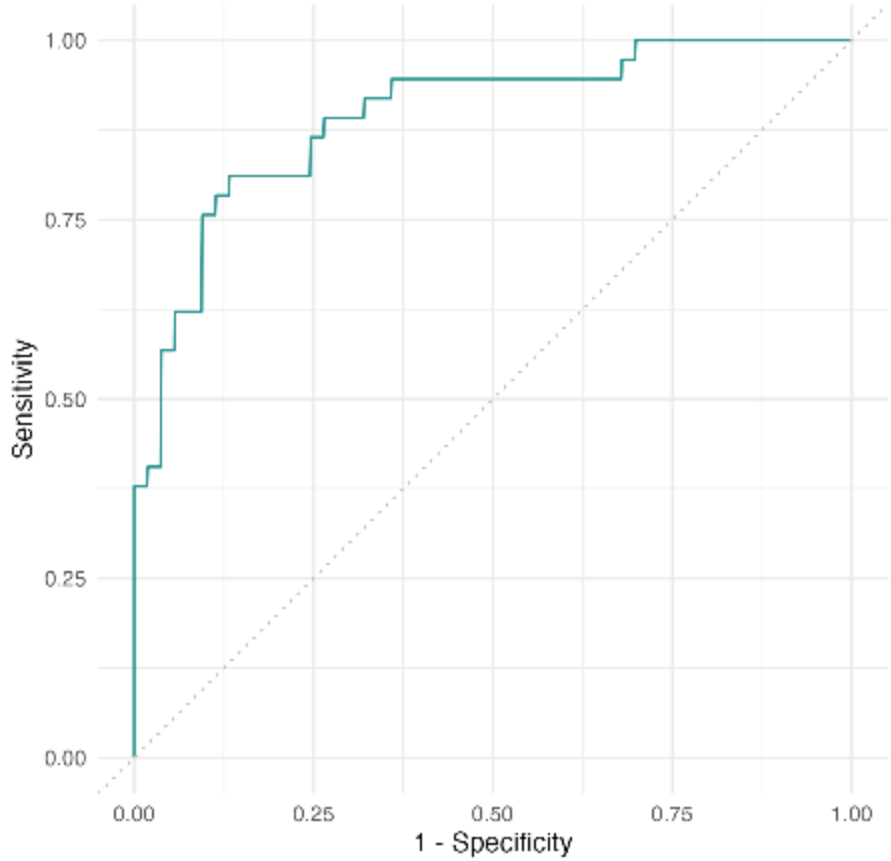
Local holdout: What does a false negative mean for a patient in this example?

Five measures, five questions

Measure	Question / denominator	Local result
Sensitivity	Among disease present: 30 / 37	81.1%
Specificity	Among disease absent: 45 / 53	84.9%
Positive predictive value	Among predicted present: 30 / 38	78.9%
Negative predictive value	Among predicted absent: 45 / 52	86.5%
Accuracy	Among all patients: 75 / 90	83.3%

Predictive values depend on the frequency of disease in the population being assessed.

ROC and AUC describe ranking



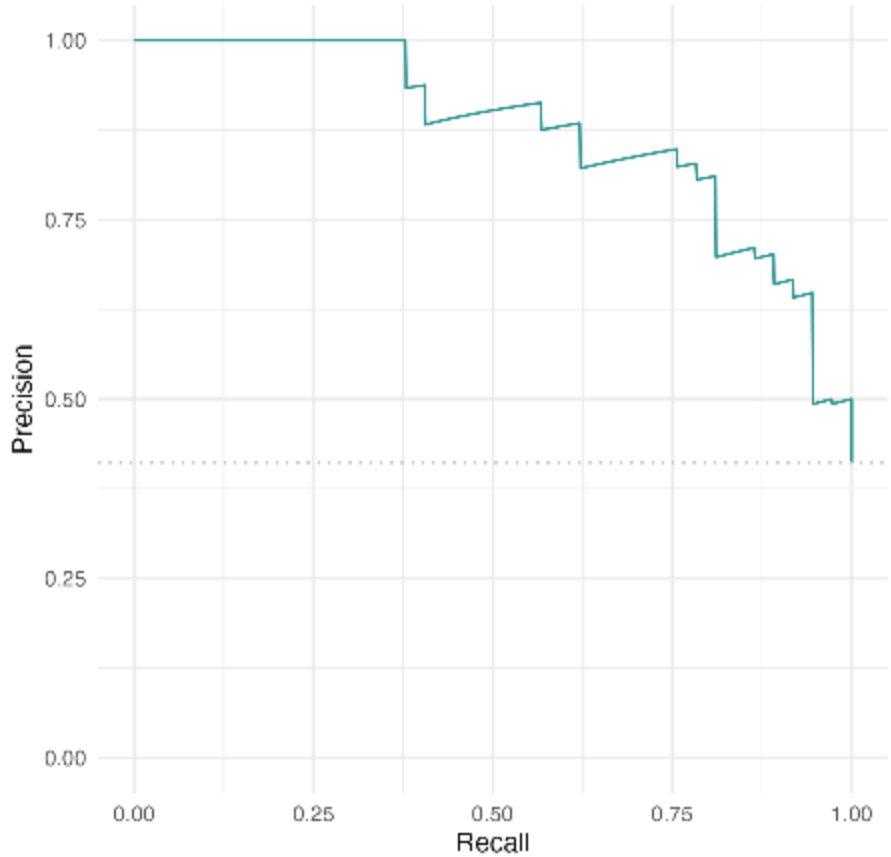
ROC compares sensitivity with false-positive rate.

False-positive rate = 1 − specificity.

AUC summarises ranking across thresholds.

AUC alone does not choose a threshold or establish clinical usefulness.

Precision–recall focuses on positive predictions



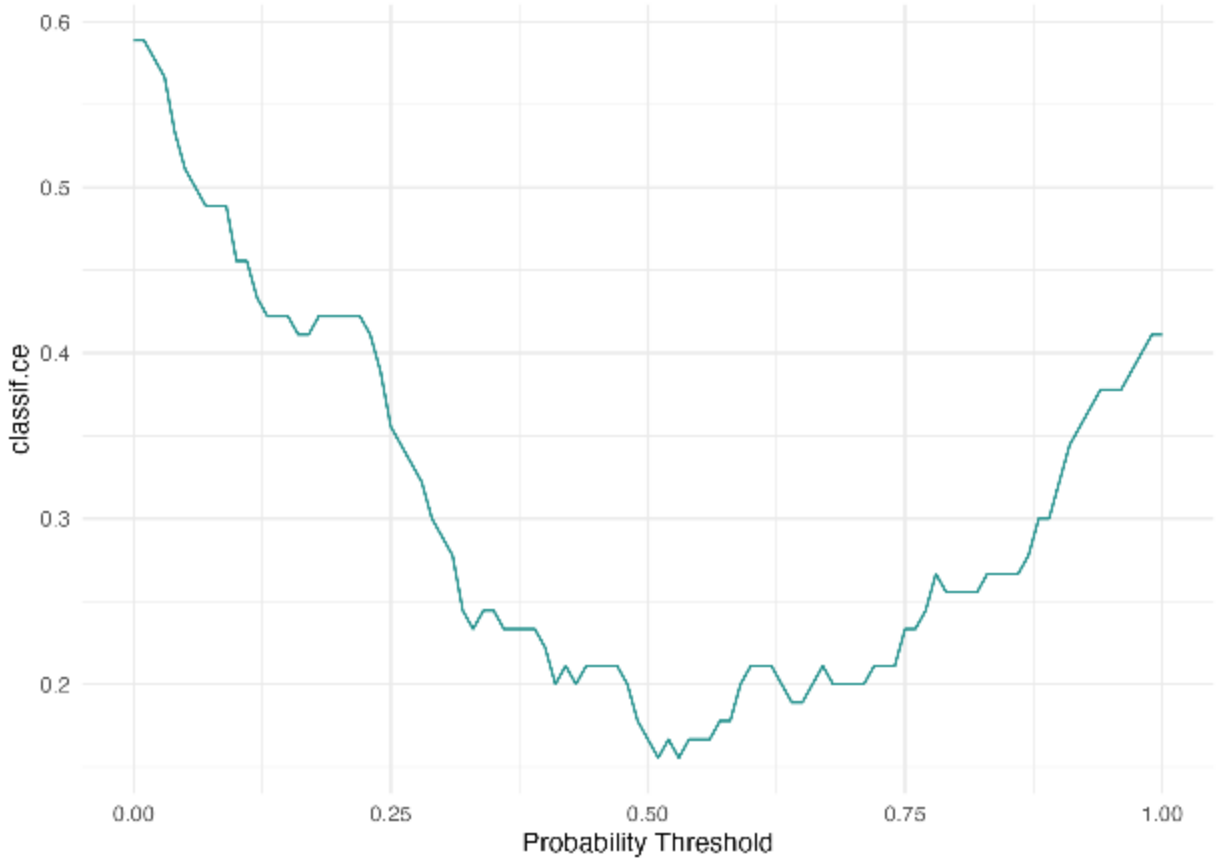
Recall = sensitivity.

Precision = positive predictive value.

The curve shows their tradeoff across thresholds.

Disease frequency matters when interpreting precision and comparing datasets.

Probability thresholds change the errors



Lower threshold usually raises sensitivity and reduces specificity.

Higher threshold usually reduces sensitivity and raises specificity.

Checkpoint: How would lowering the threshold affect missed disease and false alarms?

Workflow and limits

Define the patient, predictors and existing-disease outcome.

Fit models using training data and control complexity.

Evaluate held-out probabilities and explain the errors.

Check uncertainty, population differences and data quality.

These teaching results establish neither clinical validation nor causation.

Appendix: resampling

Cross-validation rotates which fold stays out of fitting.

Repeated subsampling draws different training/test splits.

Bootstrap samples training rows with replacement.

Completed locally: 3-fold CV, 100 subsamples and 40 bootstrap repetitions.

Appendix: tuning and nested cross-validation

A hyperparameter controls how a model learns.

Inner resampling selects settings within training data.

Outer folds assess the whole selection procedure.

Full grid search and full-budget nested CV remain unexecuted. Only reduced smoke runs passed.

Appendix: interpretation

Approach	Question
Feature importance	Which predictors contribute to this fitted model?
Feature effects	How do predictions vary with a predictor?
Local explanation	What helps explain this one prediction?

Conceptual extension. No computed feature ranking or local explanation is supplied.

Appendix: synthetic observations

A generator learns patterns from the observed data.

ARF fits the generator; FORDE estimates density.

FORGE samples 270 synthetic observations.

Generated rows are synthetic examples. They are not additional observed patients.

Appendix: sources and reproducibility

Wright et al. (2026), Machine Learning in Epidemiology
arXiv:2602.16352v1, sections 2–6

OpenML heart-statlog, dataset 53, version 1

Local notebook, scripts, saved figures and output records

Source details, timing and evidence gaps accompany this deck.